

PARWAAZ

*An AI-based Performing Arts Coach, for you, right there,
All at ONCE*

GROUP 10

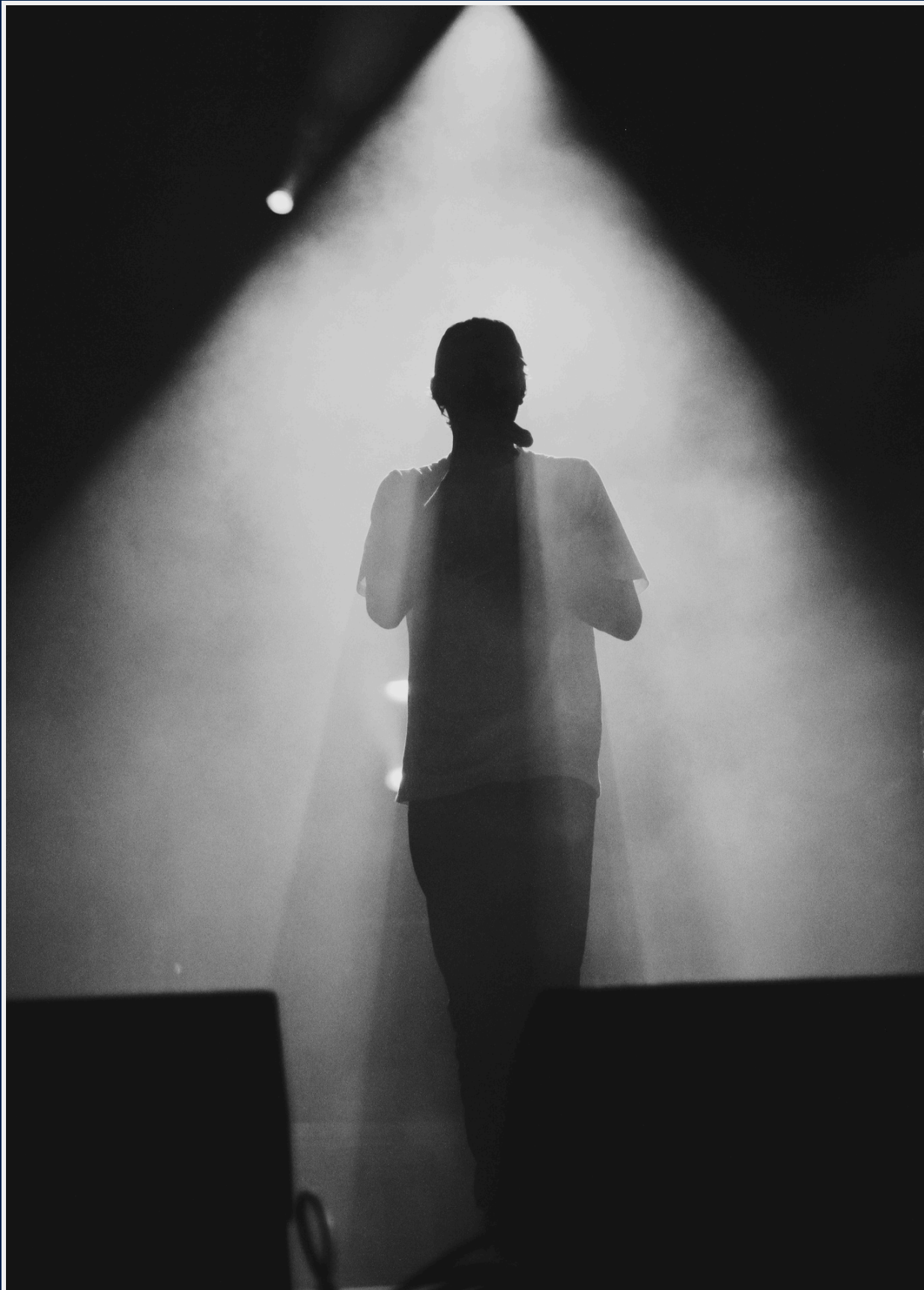
Khushi
Pratima
Vanshika
Shivani
Sejal



PROBLEM?

Aspiring performers in India face two key challenges:

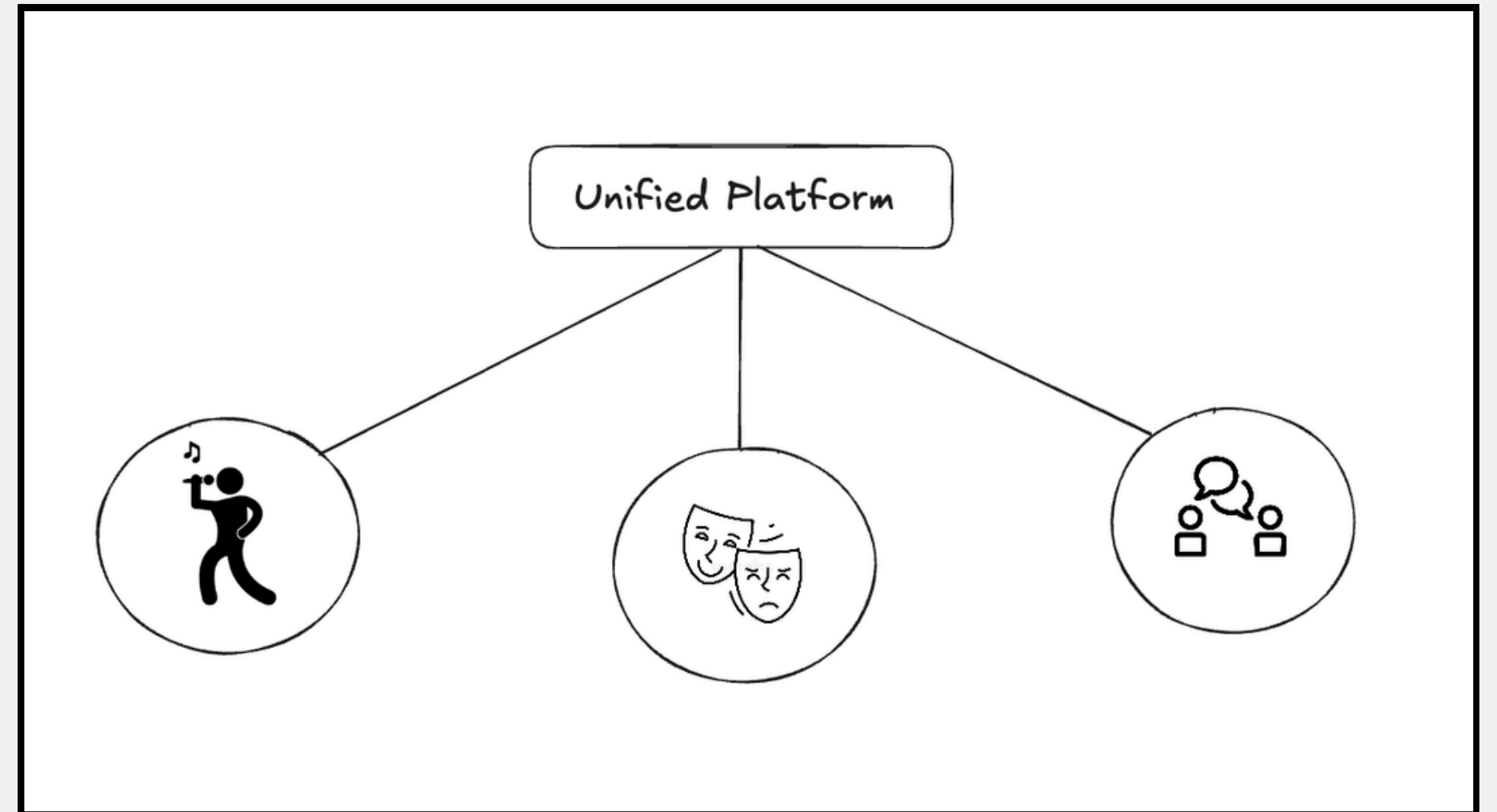
- Professional coaching is expensive and often intimidating for beginners.
- Great performance requires mastery of acting (facial expressions), speech (clarity and fluency), and singing (pitch and rhythm). However, existing AI tools evaluate only one area at a time. There is no single system that provides integrated feedback on all three from one recording, leaving performers without a complete performance assessment.



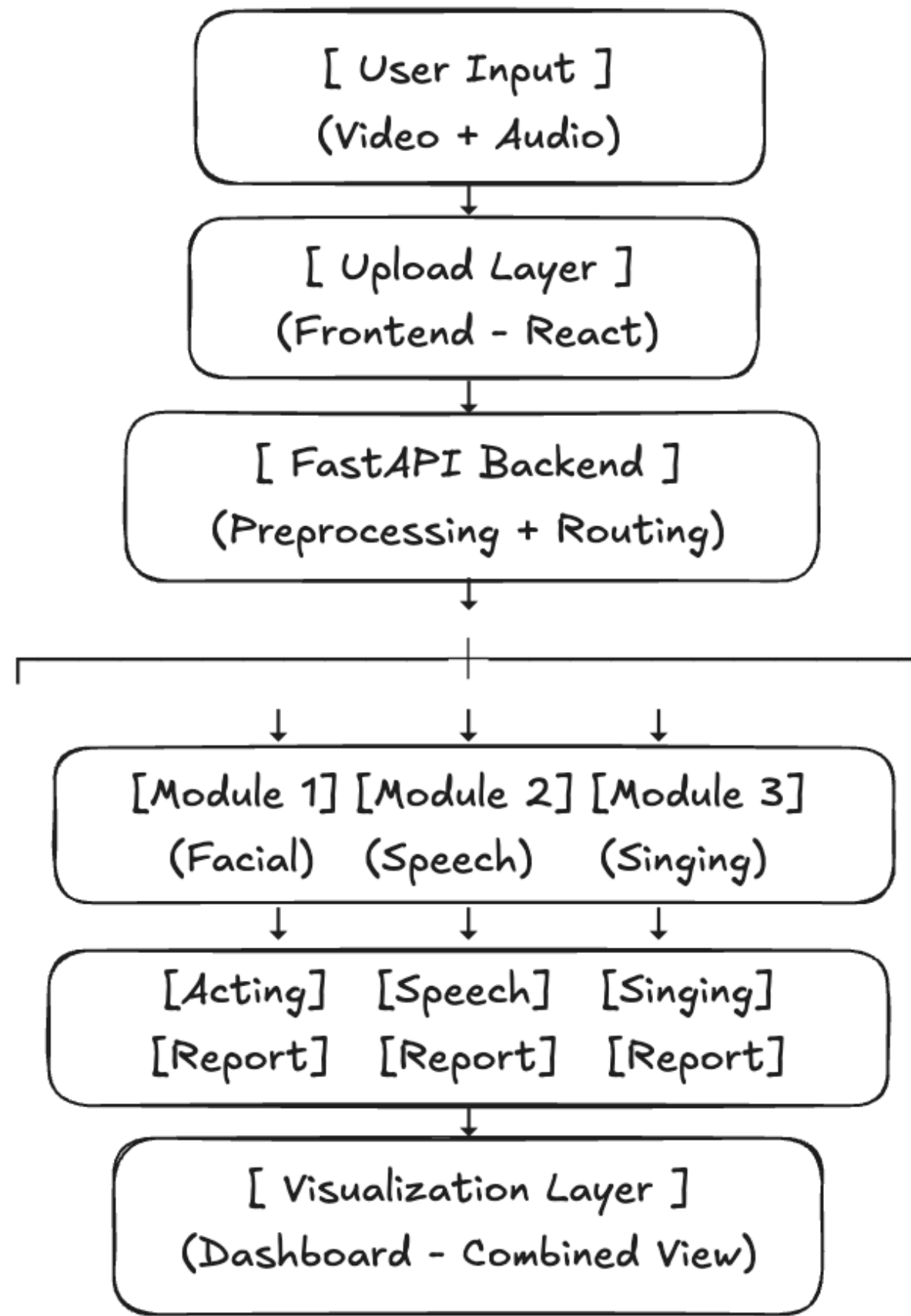
**ONE
SYSTEM.
THREE
SKILLS.**



**OUR
SOLUTION**



1. A Unified System for Multidimensional Performance Assessment
2. An AI-driven framework that analyzes visual and audio cues simultaneously.
3. Provides objective, structured feedback to support continuous improvement.



REACT.js

FASTapi

postgresql

SYSTEM ARCHITECTURE

MODULE 1 (FACIAL EXPRESSION)

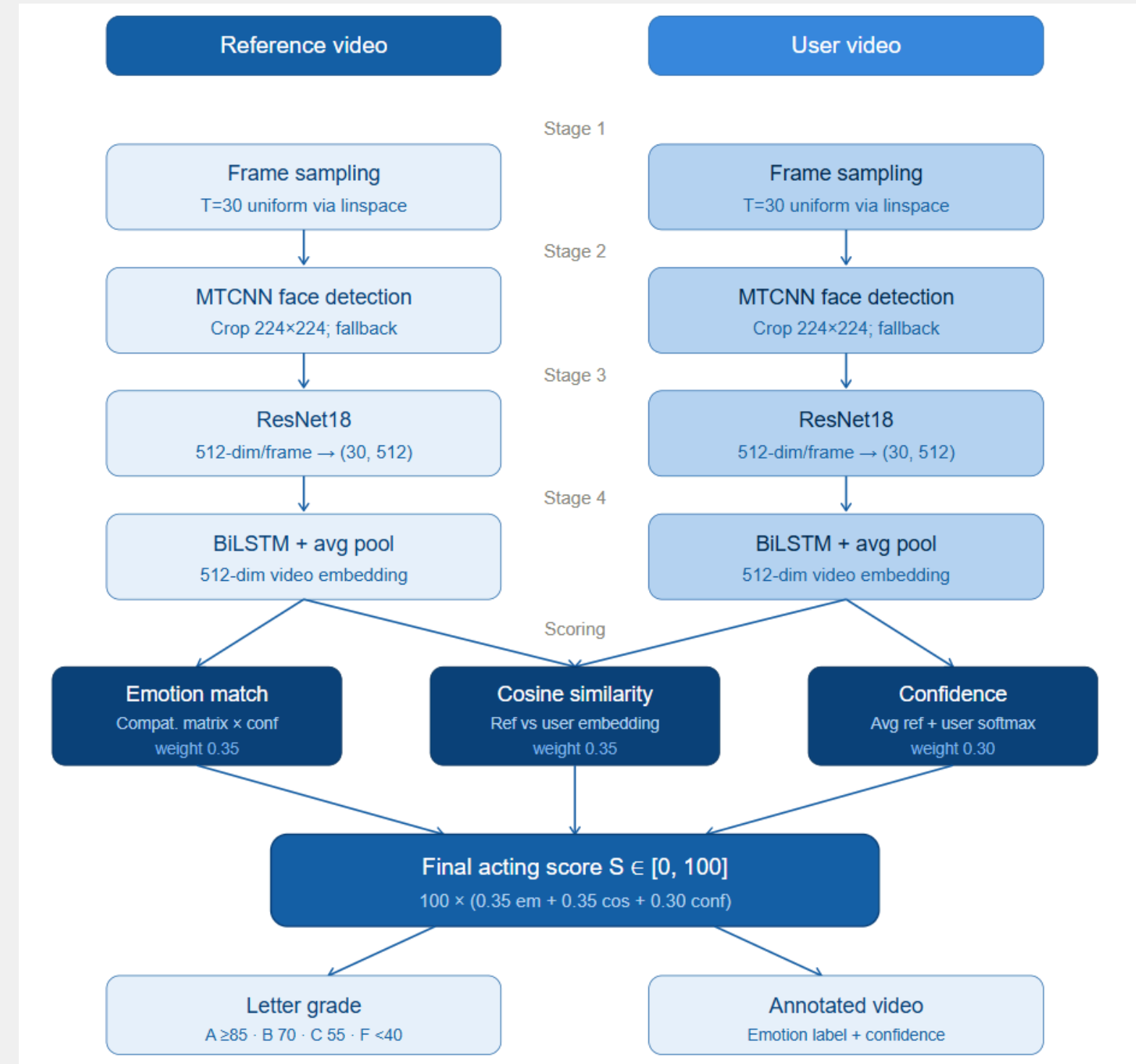
- Compares user vs reference performance
- Tracks emotion over time
- Outputs score + segment feedback

TECH:

CNN + Temporal Transformer
Cosine similarity of 512-dim embeddings

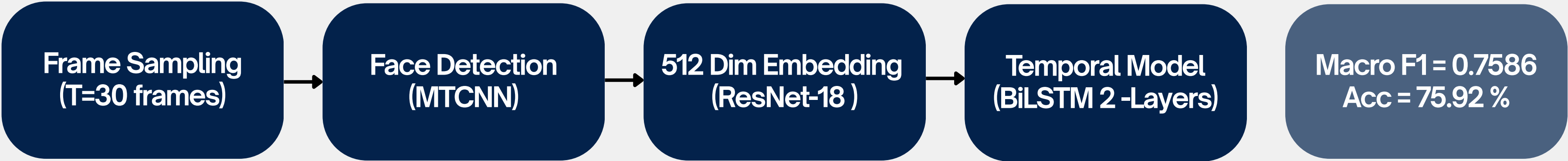
WHY THIS DESIGN?

Because acting is not a frame — it's a sequence of emotions



MODULE 1 (FACIAL ACTING AND TEMPORAL EVALUATION)

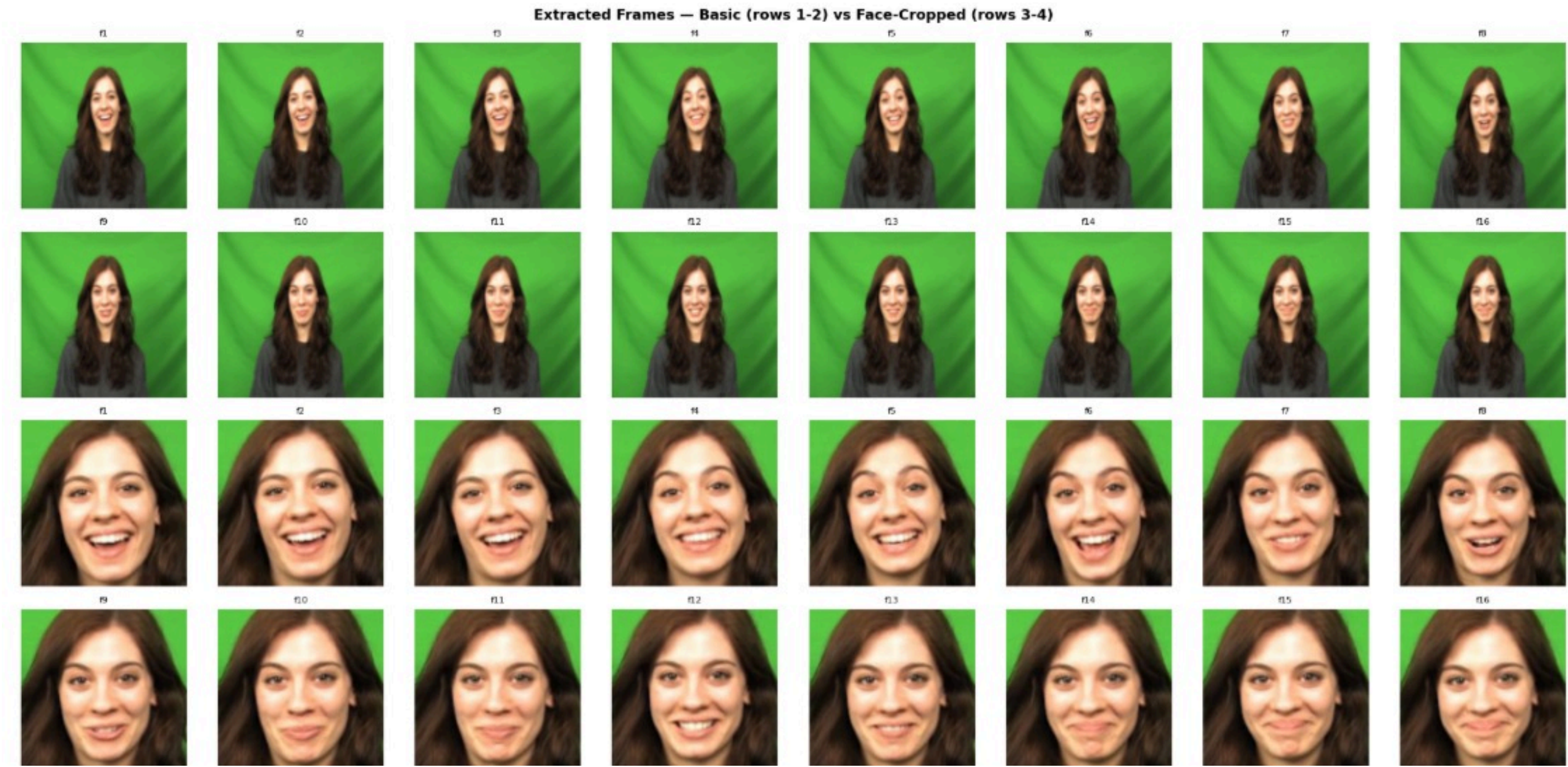
Resnet18 + MTCNN + BiLSTM | Trained on CREMA-D [7,442 clips, 91 actors, 6 emotions]



5-Component Acting Score :

Emotion Match	35%	Compatibility matrix × confidence
Distribution Score	20%	KL-divergence of per-frame softmax
Embedding Cosine	20%	512-dim BiLSTM video embeddings
Temporal Consistency	15%	Inverted std dev across 30 frames
Model Confidence	10%	Avg softmax confidence score

UNIFORM FRAME EXTRACTION FOR A SINGLE EMOTION



MODULE 2 (SPEECH ANALYSIS)

- Checks pronunciation, fluency, clarity
- Detects pauses, fillers, speed

Tech:

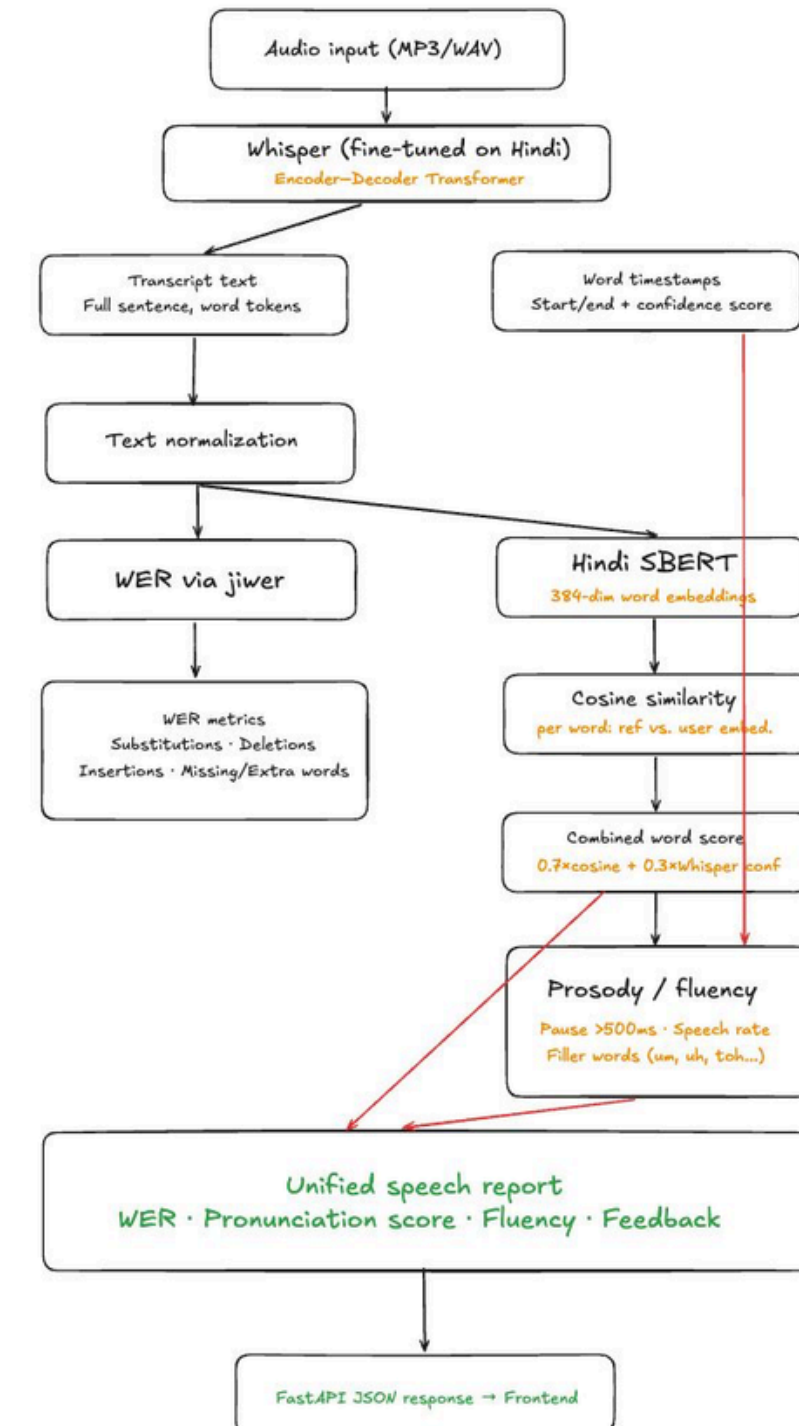
Whisper (speech-to-text)

Sentence Transformers (meaning similarity)

WHY combine both?

Whisper → “what was said”

Transformer → “how well it was said”



MODULE 2 (SPEECH ANALYSIS)

- **Fine-Tuning of Whisper— Decoder-only fine-tuning on AI4Bharat Kathbath Hindi dataset (~94k clips) using transfer learning.**

Feature	all-MiniLM-L6-v2	l3cube-pune/hindi-sentence-similarity-sbert
Type	General SBERT model	Hindi-specific SBERT model
Training Data	Mostly English text corpora	Hindi language datasets
Main Purpose	General semantic similarity	Hindi semantic + phonetic similarity
Hindi Understanding	Limited	Strong Hindi understanding

FINE TUNING

Training used:

- Cross-Entropy Loss
- AdamW Optimizer
- Learning rate: $1 \cdot 10^{-5}$
- Warmup + linear learning rate decay scheduler
- Model performance was evaluated using Word Error Rate (WER):

$$WER = \frac{S + D + I}{N}$$

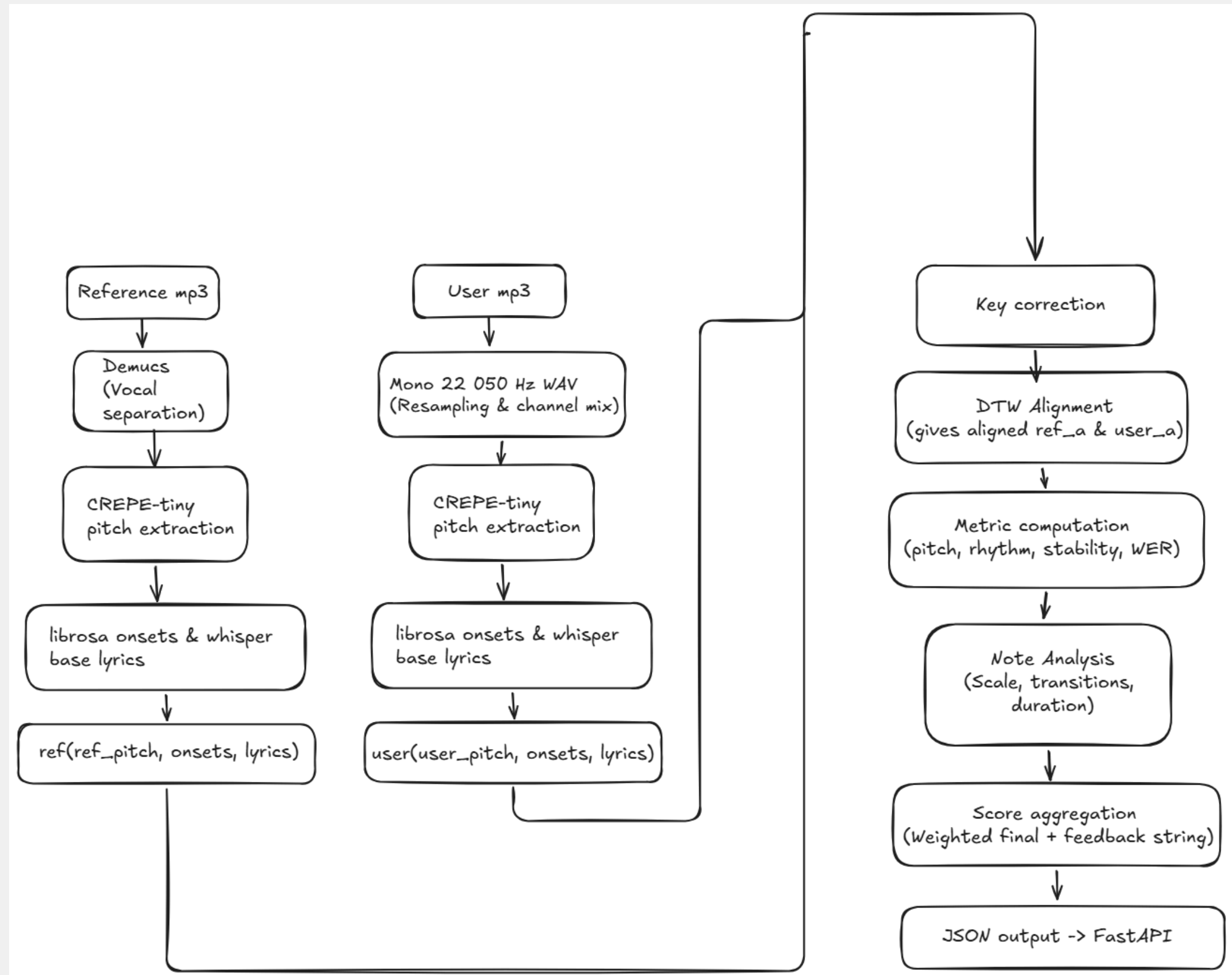
MODULE 3 (SINGING ANALYSIS)

Design Philosophy: Reference-based evaluation — no genre or language specific training needed. Works for Bollywood, Classical, Pop, Folk.

Key Technologies:

- Demucs htdemucs : Hybrid Transformer-Conv model for vocal separation (trained on MUSDB18)
- CREPE-tiny : 6-layer CNN pitch estimator, 360 bins (C1-B6), ~400K parameters
- Dynamic Time Warping : Non-linear tempo alignment via librosa
- Whisper-base : Language-agnostic lyrics transcription & WER scoring

Output: Pitch score, Rhythm score, Stability, Lyrics score → Weighted final score + detected scale + note transitions + per-segment feedback



Your Audio

▶ 0:00 / 0:46

🔊 ⋮

Live Note Tracking

⏸ 0:06 / 0:46

🔊 ⋮

Current note: E

F#

G#

F

D#

D

G

E

C

C#

B

A#

A

Notes used in this clip (shown once): highlighted live during playback.

DEMO.

Scores 🔗

Pitch Accuracy: 71.97%

Final Score: 52.94

Detected Scale

C Major

Overall estimated key/scale from the detected sung notes across the clip.

Note Transitions

- ▼ [
- 0 : "F → F#"

1 : "F# → G"

2 : "G → G#"

3 : "G# → G"

4 : "G → G#"

THANK YOU